

Solution of exercise 4, 5, 6, and 7 can be found on:
<https://www.youtube.com/@maamarbenbachir/> (Integer part of real number playlist).
You can also check Extremums playlist for the last exercise and for some useful ideas.

1. Show that, for any x, y, u, v in $\mathbb{R}$

- $|x| + |y| \leq |x + y| + |x - y|$, $\frac{|x+y|}{1+|x+y|} \leq \frac{|x|}{1+|x|} + \frac{|y|}{1+|y|}$.
- $||x - u| - |y - v|| \leq |x - y| + |u - v|$.
- $1 + |xy - 1| \leq (1 + |x - 1|)(1 + |y - 1|)$.

2. Let x, y be two real numbers, show that

- $\max(x, -x) = |x|$ • $\max(x, y) = \frac{1}{2}(x + y + |x - y|)$
- $\min(x, y) = \frac{1}{2}(x + y - |x - y|)$ • $\max(|x|, |y|) \left| \frac{x}{|x|} - \frac{y}{|y|} \right| \leq 2|x - y|$, with $x \neq 0$

3. Show that

- (a) For all $x \in \mathbb{R}$ $E(x + 1) = E(x) + 1$.
- (b) For all $(x, y) \in \mathbb{R}^2, x \leq y \Rightarrow [x] \leq [y]$.
- (c) For all $x \in \mathbb{R}$ $0 \leq [2x] - 2[x] \leq 1$.
- (d) Show that, for every $n \in \mathbb{Z}^+$ and every $x \in \mathbb{R}$, we have: $0 \leq [nx] - n[x] \leq n - 1$.

4. Show that, for every $n \in \mathbb{Z}^+$ and every $x \in \mathbb{R}$,

$$\left\lfloor \frac{[nx]}{n} \right\rfloor = [x]$$
$$\sum_{k=0}^{n-1} \left\lfloor x + \frac{k}{n} \right\rfloor = [nx]$$

5. Solve in $\mathbb{R}$ the following equation

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+1}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = [x] + 1.$$

6. Let $n \in \mathbb{N}$ and $(x_1, x_2, \dots, x_n) \in [-1, 1]^n$ such that $x_1 + x_2 + \dots + x_n = 0$. Prove that

$$\left| \sum_{k=1}^n kx_k \right| \leq E\left(\frac{n^2}{4}\right).$$

7. Calculate

$$S = \sum_{k=1}^{100} E(\sqrt{k})$$

8. For any $a, b \in \mathbb{R}$, express

$$\sup(a, b) - \inf(a, b) \quad \text{and} \quad \sup(a, b) + \inf(a, b)$$

in terms of a and b . Deduce the expression of

$$\sup(a, b) \quad \text{and} \quad \inf(a, b).$$

What about

$$\sup(-a, -b) \quad \text{and} \quad \inf(-a, -b).$$

9. Let A be a nonempty set of $\mathbb{R}$. Show that

$$\sup(-A) = -\inf(A) \quad \text{and} \quad \inf(-A) = -\sup(A),$$

where $-A = \{-x, x \in A\}$.

10. Given nonempty subsets A and B of positive real numbers. Define

$$A \cdot B := \{ab : a \in A, b \in B\} \quad \text{and} \quad \frac{1}{A} := \left\{x = \frac{1}{a} : a \in A\right\}.$$

(a) Show that

$$\sup(A \cdot B) = \sup A \cdot \sup B$$

(c) Give an example of two nonempty bounded sets A and B for which

$$\sup(A \cdot B) \neq \sup A \cdot \sup B.$$

11. Let A, B be two nonempty subsets of $\mathbb{R}$. Define

$$A + B := \{x + y : x \in A \text{ and } y \in B\},$$

and

$$A - B := \{x - y : x \in A \text{ and } y \in B\}.$$

Show that

$$\sup(A + B) = \sup A + \sup B \quad \text{and} \quad \sup(A - B) = \sup A - \inf B.$$

Establish similar formulas for $\inf(A + B)$ and $\inf(A - B)$.

12. Let be the subsets of $\mathbb{R}$ defined by

$$A = \left\{\frac{1}{2} - \frac{n}{2n+1}, n \in \mathbb{N} \cup \{0\}\right\}, B = \left\{1 + \frac{(-1)^n}{n}, n \in \mathbb{N}\right\}, C = \left\{\frac{1}{n} + (-1)^n, n \in \mathbb{N}\right\}.$$

$$D = \left\{\frac{1}{2} + \frac{n}{2n+1}, 2 - \frac{1}{n+2} : n \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}\right\}, \Omega = \left\{\frac{m+n+2mn}{m+n+mn+1} : m, n \in \mathbb{N}_0\right\}.$$

Prove that they are bounded and determine (using characterization) their bounds ($\inf$, $\sup$, $\min$, $\max$).

13. Show that

$$\sup\{x \in \mathbb{Q} : x > 0, x^2 < 2\} = \sqrt{2}.$$